MadhuTwin

A hybrid physiological–AI digital twin that predicts glycaemic adverse events in type 2 diabetes, built for India

Anish Karthik Ayyagari · Indian Institute of Technology Kharagpur

Digital Twin Challenge 2026 · Happiest Health Reimagining and Reforming Healthcare in India Summit

About the builder

AK

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Solo entrant and team leader

Physiological twin

Glucose-insulin model with Indian meals and medicines

Personalisation

EHR prior × sensor likelihood, MAP fit

Hybrid ML + validation

Leave-patients-out study on real CGMacros data

Bharat synthetic cohort

80 fictional Indian patients with full EHR and wearables

API and copilot

FastAPI service, FHIR export, grounded summaries

Clinician dashboard

React triage, live twin, what-if, AGP, evidence

India's diabetes burden is vast, and mostly invisible between clinic visits

101 M
people living with diabetes

136 M
people with prediabetes

11.4 %
national diabetes prevalence

15.3 %
prediabetes prevalence

Source: ICMR-INDIAB, Anjana et al., Lancet Diabetes & Endocrinology 2023 (113,043 adults, 31 states/UTs).

Care is reactive. A quarterly HbA1c averages three months into one number. It cannot show the post-meal spikes or the afternoon lows that drive complications and emergencies.

Sensors are arriving; decisions are not. CGMs and smartwatches now generate thousands of readings per patient, but doctors receive data dumps, not early warnings.

Two adverse events doctors rarely see in time



Post-meal spikes

- High-glycaemic-load meals (rice, refined wheat, sweets, tea with sugar) push glucose above 180–250 mg/dL for hours.
- Post-prandial excess drives a large part of the HbA1c gap and is linked to cardiovascular risk.
- The response to the same meal differs from person to person: one diet plan does not fit all.



Drug-induced hypoglycaemia

- Sulfonylureas remain widely prescribed in India because they are inexpensive and effective.
- Skipped meals, unplanned exertion, older age and CKD turn them into a leading cause of severe lows.
- Fasting days and late meals make the risk patient-specific and time-specific.

MadhuTwin: a living model of each patient, fed by their record and their sensors

Static record (EHR)

- Demographics, diagnoses
- HbA1c, fasting glucose and insulin, lipids, eGFR
- Medications, TCF7L2, polygenic risk

Live wearable streams

- CGM every 5 minutes
- Heart rate, steps, METs
- Sleep stages, HRV, meal log

Digital twin

- Personalised glucose–insulin physiology
- Indian meals and medicines
- Synced to every CGM reading
- Hybrid ML corrects the residual
- Calibrated uncertainty and reasons

Forecast 2 h ahead

- with 90 % intervals

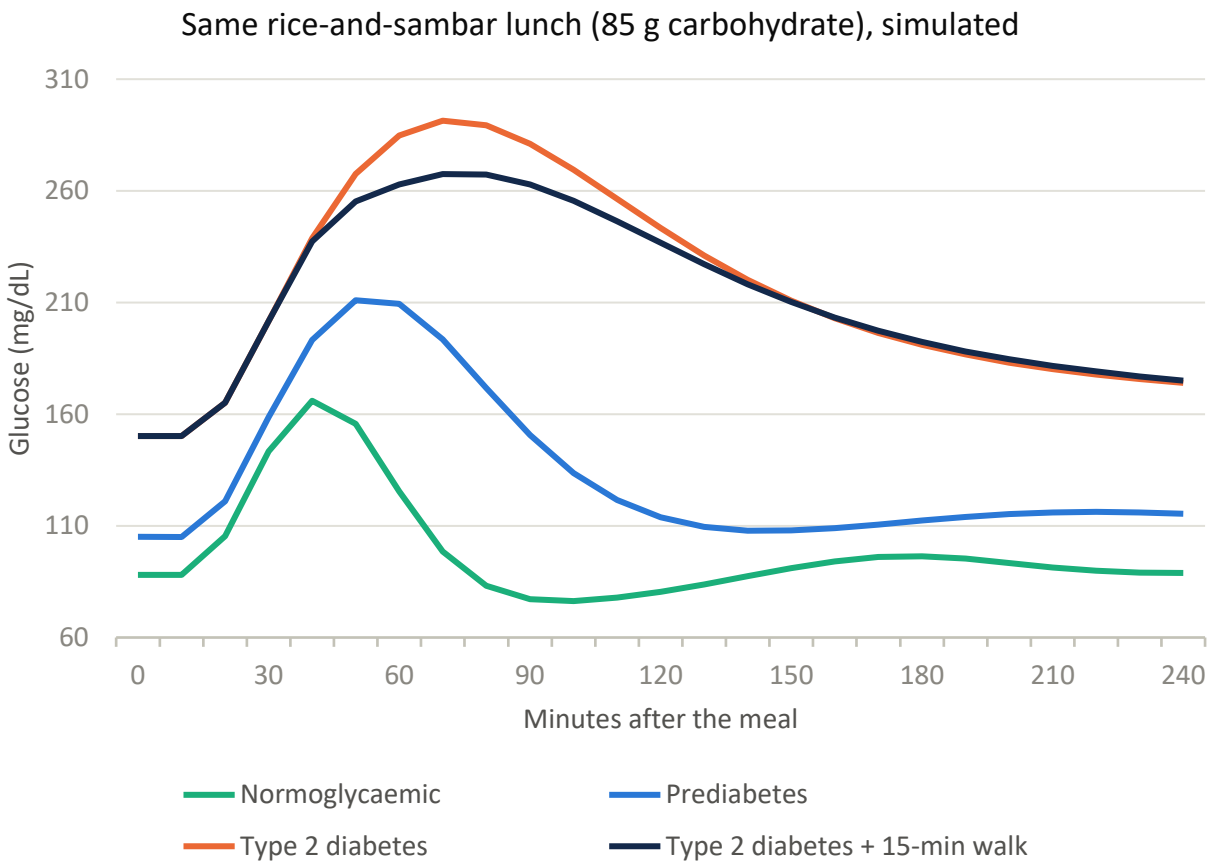
Adverse-event alerts

- hyper, spike, hypo + why

What-if simulation

- meals, walks, sleep, drugs

A physiological core that understands Indian meals and medicines



Meal absorption

Fast and slow chains; glycaemic index, fat, fibre and protein shape the curve

Insulin secretion and action

Beta-cell response, insulin sensitivity, glucose effectiveness

Activity, sleep, rhythm

METs raise uptake; short sleep lowers sensitivity; dawn and circadian effects

Medicines used in India

Metformin, sulfonylureas, DPP-4i (teneligliptin), SGLT2i, voglibose, pioglitazone, basal insulin

The record sets the prior; the sensors personalise it

1 Prior from routine labs

HbA1c, fasting glucose and insulin (HOMA-IR, HOMA-B) and weight map to priors on the patient's physiology. Useful on day zero.

2 Onboarding fit

The first days of CGM, meals and activity refine the parameters by fitting the twin's own 2-hour forecasts (MAP estimate).

3 Continuous sync

Every new CGM sample corrects the virtual patient's state; insulin, gut content and exercise effects evolve through physiology.

41.9 → 29.7

mg/dL two-hour error of the physiology alone, before vs after onboarding (real patients)

ρ = -0.68

twin insulin sensitivity vs lab HOMA-IR across 45 real participants

Output: an interpretable phenotype. Insulin sensitivity, beta-cell response, disposition index, absorption speed, exercise response and dawn effect, normally needing a clamp study, estimated from data the patient already generates.

Machine learning corrects what physiology misses, with honest uncertainty

Residual forecaster

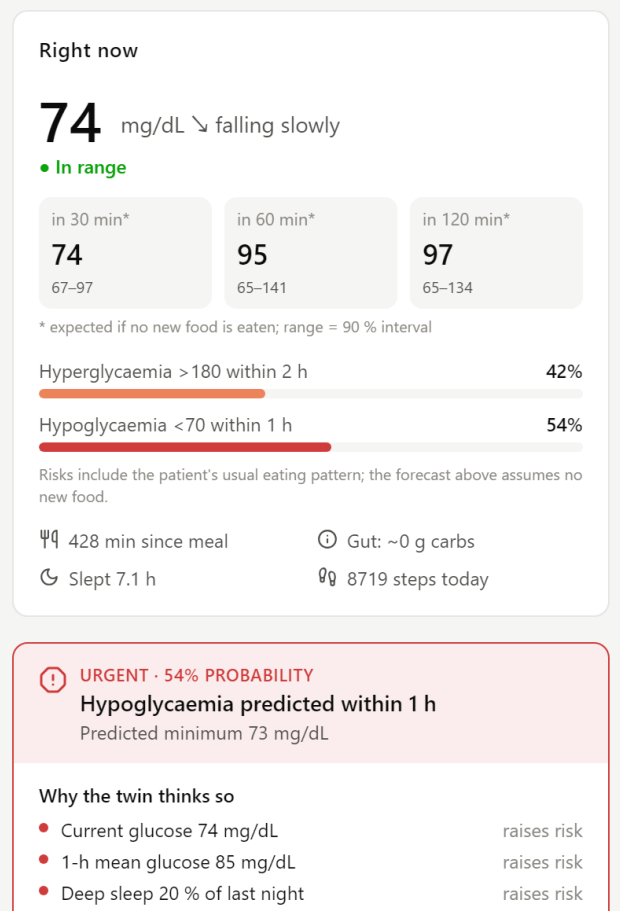
Gradient-boosted trees per horizon (30–120 min) see every stream plus the twin's own forecast, and learn when to trust physiology. Forecasts assume no new food; meals enter through the twin.

Conformal intervals

Conformalised quantile regression gives 90 % prediction intervals with a distribution-free coverage guarantee. Observed coverage on unseen patients: 91.7% at 2 h.

Explained alerts

Calibrated risks for hyperglycaemia, spikes and hypoglycaemia. Exact TreeSHAP values, grouped by data stream, tell the doctor why each alert fired.



Every on-screen prediction comes from a cross-validation model that never saw that patient.

An afternoon hypoglycaemia, flagged before it happens



Light blue after 'Now': what actually happened (revealed for the demo).

Who

Ramesh, 61, glimepiride + CKD, skips lunch and walks to the market

What the twin sees

Falling trend, recent activity, a long fast and a sulfonylurea

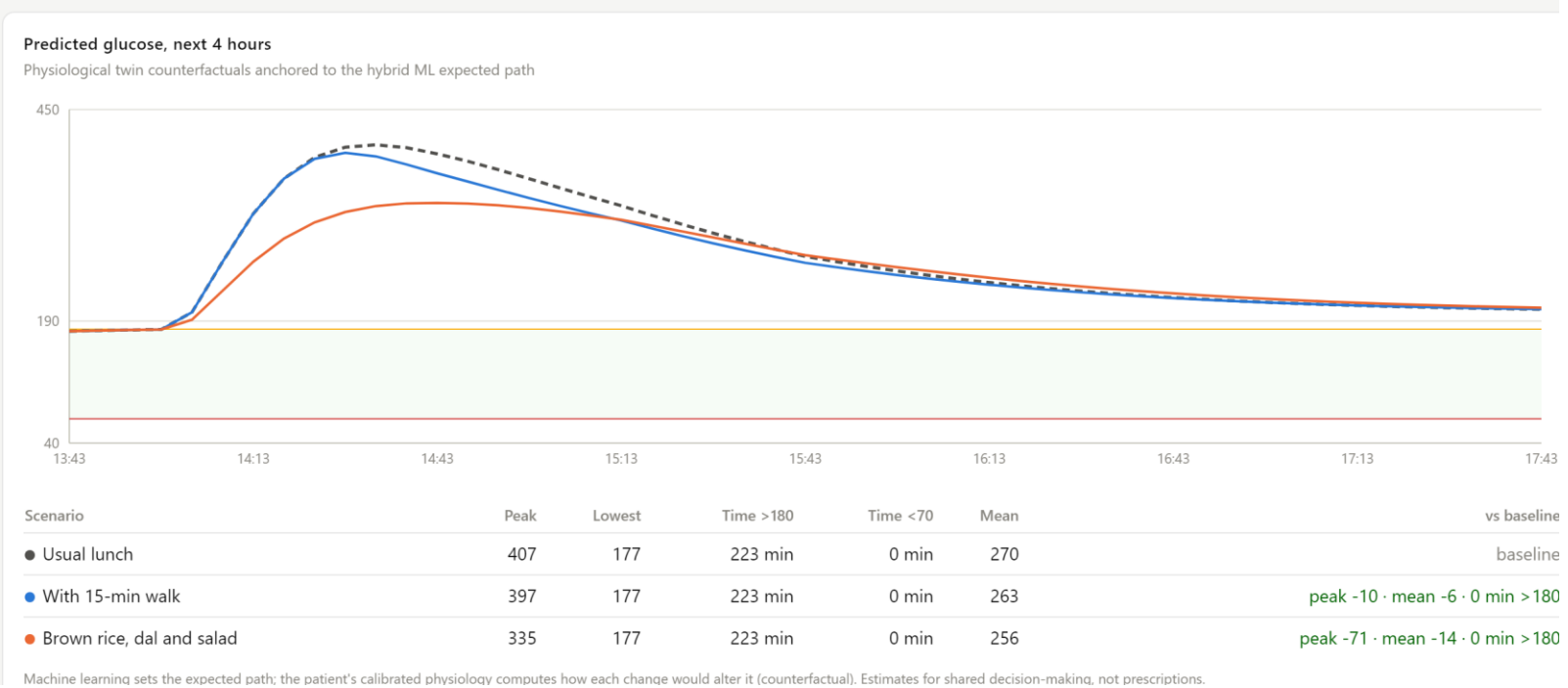
Why risk, not just a line

The median forecast expects recovery; the calibrated risk model flags a 54 % chance of a low. The low happens.

What the doctor gets

An urgent alert with reasons, a twin-tested action (15 g carbohydrate) and a prompt to review the dose

What-if: test a meal swap or a walk on the patient's own physiology



Scenario	Peak	Lowest	Time >180	Time <70	Mean	vs baseline
Usual lunch	407	177	223 min	0 min	270	baseline
With 15-min walk	397	177	223 min	0 min	263	peak -10 · mean -6 · 0 min >180
Brown rice, dal and salad	335	177	223 min	0 min	256	peak -71 · mean -14 · 0 min >180

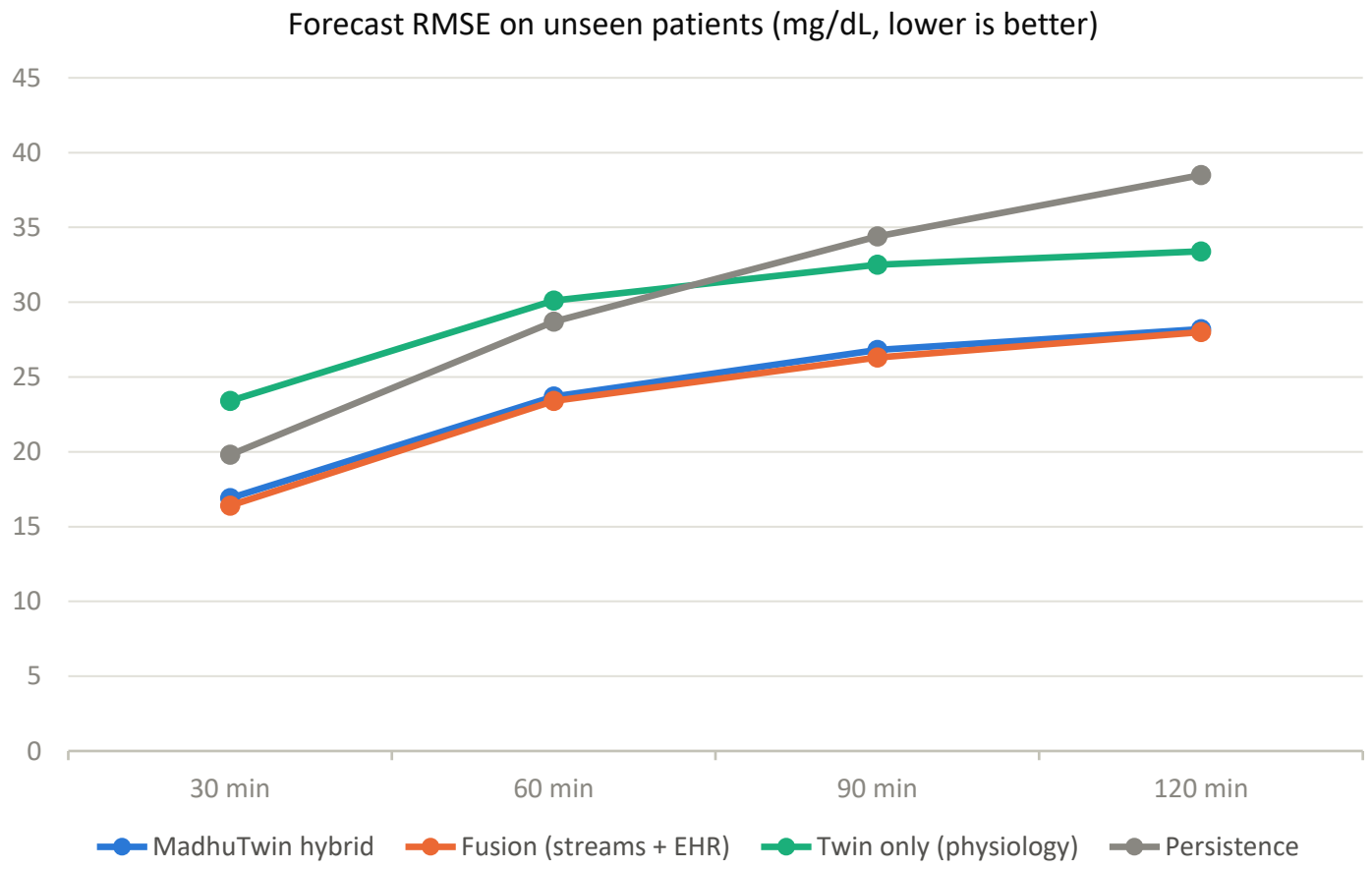
Machine learning sets the expected path; the patient's calibrated physiology computes how each change would alter it (counterfactual). Estimates for shared decision-making, not prescriptions.

Lakshmi's usual rice-and-rasam lunch vs a 15-minute walk vs brown rice with dal (twin estimates).

- Indian meal library with glycaemic index, portions and timing
- Walks, sleep and medication changes (add DPP-4i, SGLT2i, AGI; stop sulfonylurea)
- Peak, time above 180 and time below 70 compared side by side
- The twin also ranks the best next action automatically

ML sets the expected path; the patient's physiology computes the counterfactual change.

Validated on real patients: 27% lower two-hour error than persistence



28.2 mg/dL

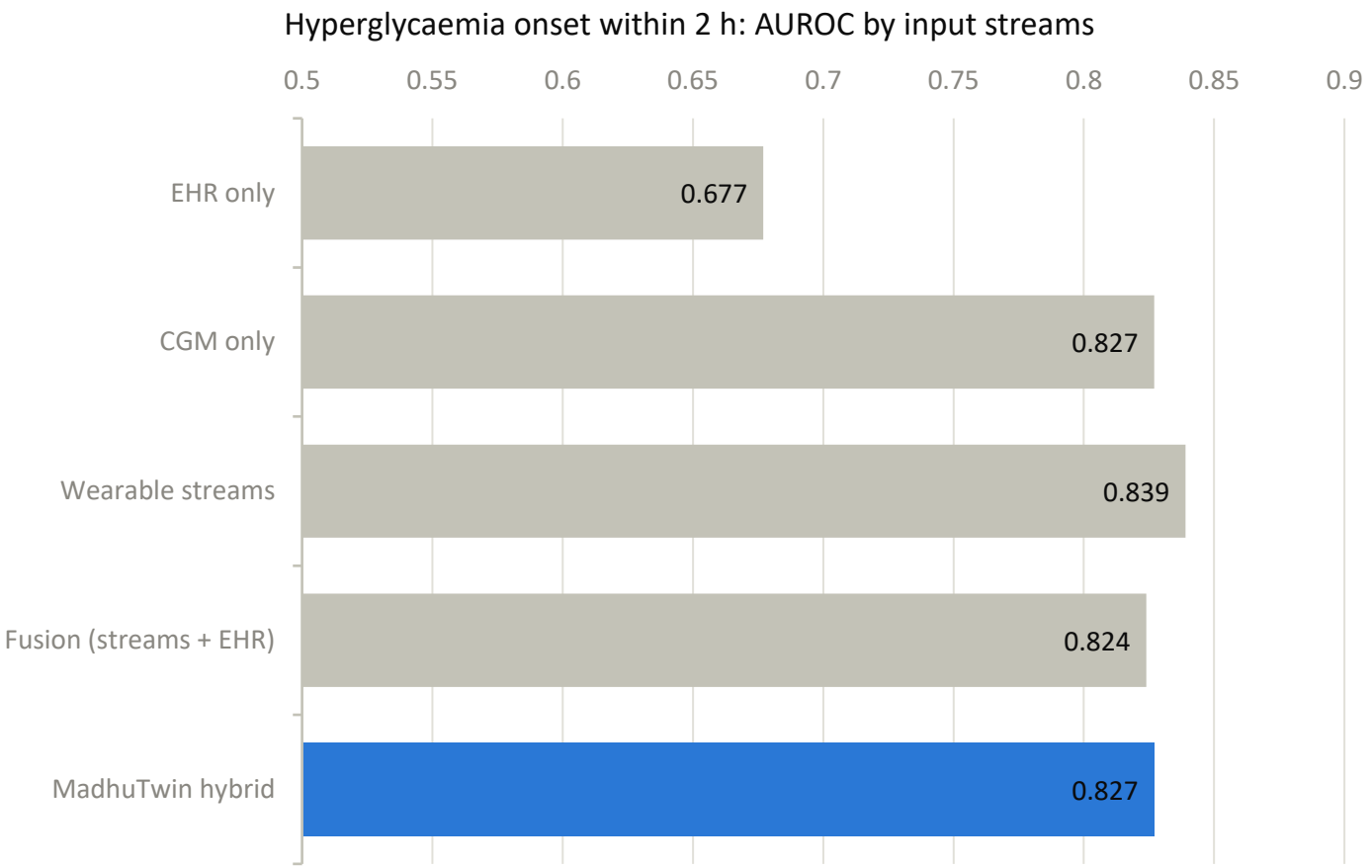
2-hour RMSE vs 38.5 for persistence

99.0 %

of 2-hour forecasts in Clarke error-grid zones A+B (clinically acceptable)

CGMacros (PhysioNet): 45 adults with CGM, Fitbit, meals and labs. 5-fold leave-patients-out; twins personalised only on each patient's first days; metrics on 311 unseen patient-days. Native sensor samples only (no interpolation leakage).

The record tells you who is at risk; the streams tell you when



75 min

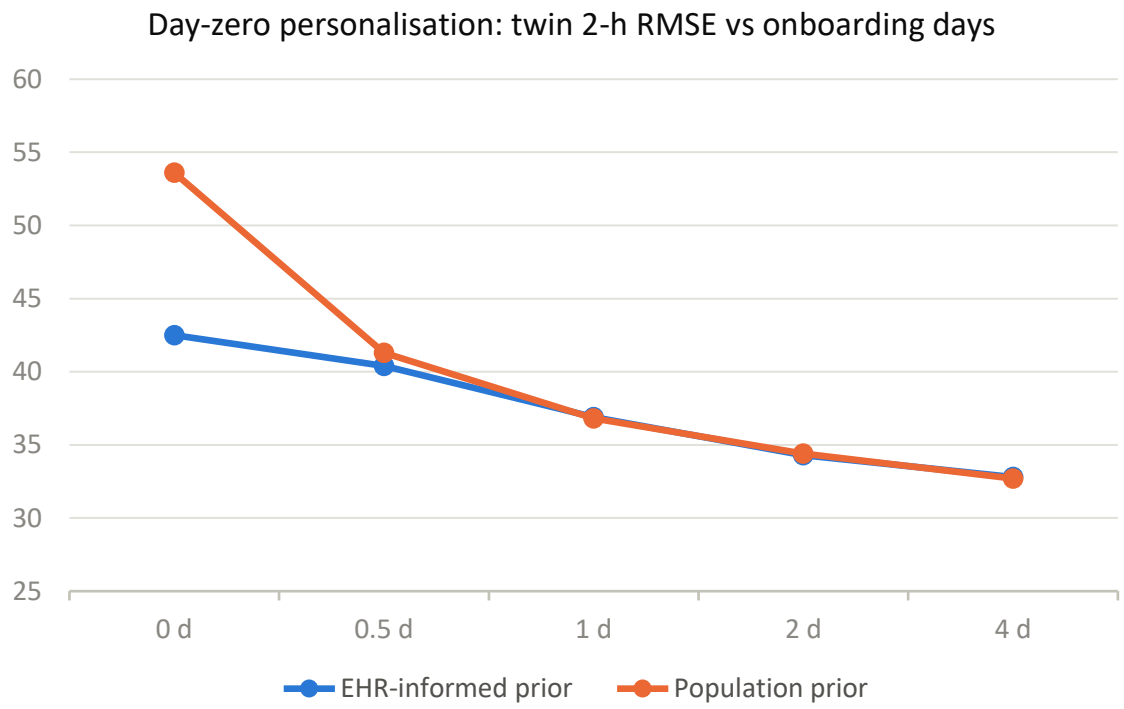
median warning before glucose crosses 180 mg/dL

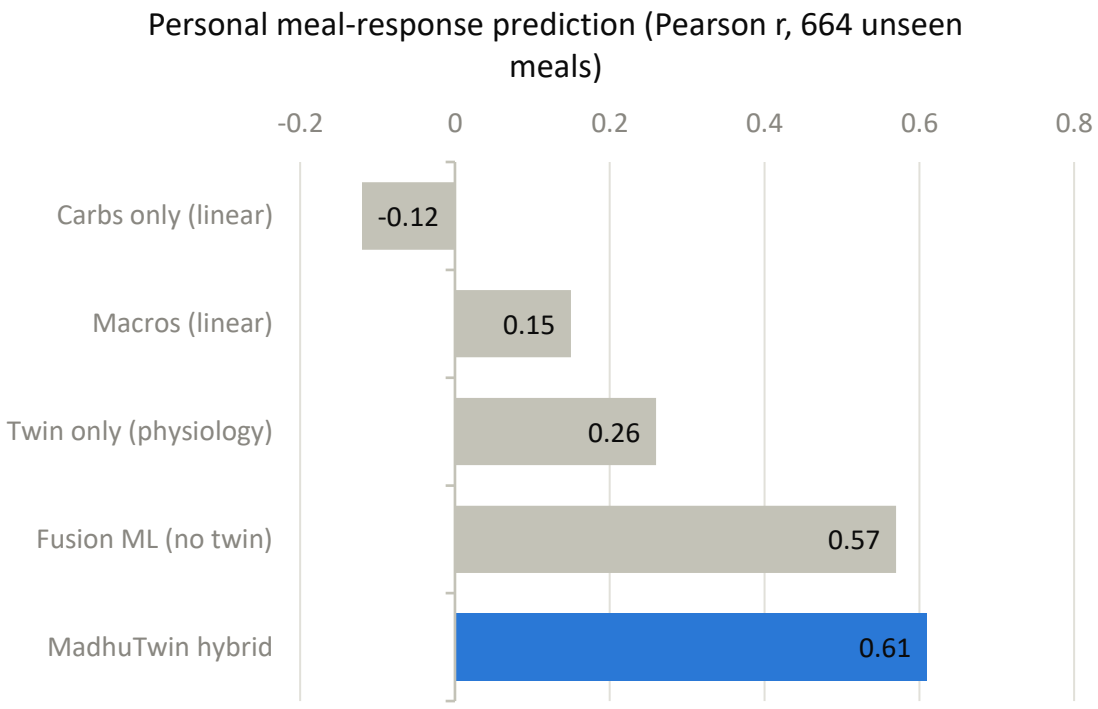
61%

of hyperglycaemic episodes flagged at least 15 minutes ahead

1.4 false alerts per patient-day at a 0.5 threshold. Spike (+50 mg/dL) AUROC 0.82. EHR alone (AUROC 0.68) identifies baseline risk; fusing it with live streams gives the best early warning.

Where the physiological twin earns its place





EHR prior: 21% lower error on day zero. The record makes the twin usable before any sensor data; after about a day the streams dominate.

Simulation and physiology. Carbohydrate counting alone does not predict an individual's response; the twin does. On the synthetic Indian cohort, 57% of hypoglycaemia episodes are flagged at least 10 min ahead.

What we found, including what did not work

Twin features vs best fusion model

On rolling real-data forecasts, adding twin features did not beat the best fusion model. Their value is counterfactual simulation, physiology, day-zero personalisation and meal-response prediction.

Three ideas rejected after validation

A disturbance observer, an activity-persistence prior and a fasting steady-state constraint were tested on held-out data and switched off: none improved accuracy.

Small, non-Indian real cohort

45 US adults, ~10 days each. Indian demographics, diets and drugs are tested on a synthetic cohort, always labelled and never mixed with real evidence.

Lows: trust the risk, not the line

Lows are rare in CGMacros (0.35 % of moments) and point forecasts regress towards typical levels before them; the calibrated hypoglycaemia classifier (AUROC 0.85 real, 0.99 synthetic) is the primary signal.

Built for India: clinically, culturally and legally

Indian context

- 54 Indian dishes with glycaemic index (IFCT 2017)
- Indian prescribing: teneligliptin, voglibose, glimepiride, gliclazide
- Asian-Indian BMI and waist cut-offs; IDRS
- Patient language recorded for notices and nudges

Privacy by design (DPDP Act 2023)

- Purpose-bound consent per patient
- Data minimisation and retention limits
- Only open de-identified or fictional data used
- Decision support; no dosing instructions

Interoperable

- HL7 FHIR R4 bundles (ABDM-compatible structure)
- LOINC, SNOMED CT, ICD-10, WHO ATC codes
- HL7 CGM IG summary codes; AGP report
- Open source: Python, FastAPI, React

From prototype to clinic

Now

Validated prototype

Real-data evidence, Indian synthetic cohort, clinician dashboard, FHIR export

Next 6 months

Clinical pilot

Remote-monitoring clinic in Bengaluru: alert-guided vs usual care on time in range and hypoglycaemia

12 months

Scale features

Photo-based Indian meal logging; low-cost sensor support; on-device twin for low connectivity

Beyond

More conditions

Same twin framework for hypertension and cardiovascular risk; CDSCO SaMD pathway

Who benefits: endocrinologists and physicians running remote-monitoring clinics, diabetes educators, and community programmes where every specialist minute counts.



Warn early. Explain why. Test before acting.

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